



**KENYATTA UNIVERSITY
DEPARTMENT OF GEOGRAPHY**

**AGE 210: QUALITATIVE AND QUANTITATIVE TECHNIQUES IN
GEOGRAPHY**

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STATISTICS

LESSON 1: PROBABILITY STATISTICS

1.1 Introduction

In this lesson the theory, basic concepts and types of probability distributions are discussed bearing in mind the geographical application of probability statistics. Often geographers are confronted with such questions as what is the frequency of certain events likely to occur at given intervals? What is the frequency of a given observed phenomenon exceeding the average? For instance, what is the frequency of a given amount of rainfall in a given time interval exceed the average? To answer the above questions geographers must turn to probability statistics.

1.2 Objectives

By the end of this lesson the learner should be able to:

- Define probability and explain the laws of addition and multiplication
- Explain and compare types of probability distributions
- Apply the knowledge of probability distributions in solving statistical problems in geography

1.3 Probability Theory

Probability can be described as the relative frequency with which an event occurs or will occur in the long run. Probability is measured on a scale ranging from 0 (meaning impossible) to 1 (meaning certain).

DARREN GREGORY THE 9TH DELEGATE

Statistics and

Cartography

5

Note:

If a fair coin is tossed 100 times, It will turn up heads about 50 times and tails about 50 times. Therefore the probability of it turning up heads is $50/100 = 0.5$. The probability of it turning up tails is also 0.5. Thus: $P(\text{heads}) = P(\text{tails}) = 0.5$.

$P(\text{dying}) = 1$ (i.e. 100 out of 100 people will die- death is certain). $P(\text{cat mating with dog}) = 0$, (i.e. a cat has never mated with a dog, is never likely to do it, i.e., it is impossible)

It is important to understand the following concerning probability:

- (a) The knowledge of how often an event will occur in the long run may give little guidance as to when it will occur in the short run. If you have won the toss five times

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in a row, you are just as likely to win it again next time as if you had lost five times in a row.

- (b) If the probability of an event occurring is known, then the longer the run, the more accurately one can predict the relative frequency or proportion of heads and tails from 1000 tosses of a coin can be predicted to be near 50:50 with greater confidence than from 100 tosses. Which is similarly more predictable than from 10.

Question:

1. If a fair dice is thrown 600 times about how many times will a six be thrown? What is the probability of throwing a six?
2. About 106 boys are born to every 100 girls.
 - (a) What is the probability that the next child born will be a boy?
 - (b) What is the probability that it will be a girl?
 - (c) What is the probability that it will be either a boy or a girl?

1.3.1 Law of Addition

If a dice is thrown 600 times, a one will be thrown about 100 times, a two 100 times and so on. That is either a one or a two will be thrown 200 times, i.e. $p(\text{one or two}) = 100/600 + 100/600 = 1/6 + 1/6 = 1/3$. That is the probability of either of two events occurring is obtained by adding the probabilities of the events occurring individually.

Example:

From a study of a farm's records, it is found that over the past 100 years, heavy crop losses were suffered as follows:

In 5 years as the result of spring frost, in 4 years as the result of summer drought and in 7 years as the result of autumn storms. Assuming no overall change in conditions, the probability of heavy crop losses from any of these causes is calculated as follows: **Note:**

$$P(\text{spring frost damage}) = 5/100 = 0.05$$

$$P(\text{summer drought damage}) = 4/100 = 0.04$$

$$P(\text{autumn storm damage}) = 7/100 = 0.07$$

If it is further assumed that there is no connection between the incidence of the three types of weather, $P(\text{spring frost, or summer drought or autumn storm damage}) = 0.05 + 0.04 + 0.07 = 0.16$. In other words, heavy crop losses from any one of the above weather types can be expected on about 16 occasions in 100 years, although the data gives no indication of when these will occur.

1.3.2 Law of Multiplication

If two dice are thrown together, there are 36 possible outcomes, thus:

6,6	6,5	6,4	6,3	6,2	6,1
5,6	5,5	5,4	5,3	5,2	5,1
4,6	4,5	4,4	4,3	4,2	4,1
3,6	3,5	3,4	3,3	3,2	3,1
2,6	2,5	2,4	2,3	2,2	2,1
1,6	1,5	1,4	1,3	1,2	1,1

Each of these outcomes is equally probable if the two dice are fair. The probability of throwing a double six is $1/36$ or 0.028. This calculated as follows:

$$P(\text{six}) = 1/6 \text{ and } P(\text{double six}) = 1/6 \times 1/6 = 1/36$$

Therefore the probability of two or more events occurring together (either simultaneously or in sequence) is obtained by multiplying the probabilities of the events occurring individually. For example, if the probability of heavy crop loss in a year is 0.16, the probability of heavy crop losses being suffered in two successive years is given by multiplying together the probabilities of heavy crop losses in each of the years individually. Thus: $P(\text{heavy loss in successive years}) = 0.16 \times 0.16 = 0.0256$. (It is assumed that conditions in one year do not affect conditions another year).

We can say then that the addition law applies to **either/or situations** and the multiplication law to **and situations**.

Activity:

- 1 Throw two dice simultaneously and find the probability of a double being thrown?

- 2 What is the probability of throwing two dice so that the sum of their score is (a) 7, (b) 10 (c) more than 8?

1.4 Probability Distribution

The table below shows the possible outcomes of throwing two dice. If the pairs of numbers are summed, the results are as follows:

12	11	10	9	8	7
11	10	9	8	7	6
10	9	8	7	6	5
9	8	7	6	5	4
8	7	6	5	4	3
7	6	5	4	3	2

From the table, we can see that there is only one way of getting a score of 12, two of getting 11 (i.e. 5 and 6 or 6 and 5), three of getting 10, and so on.

We know that the outcome of throwing two dice has 36 equal probability of occurring. Thus: $P(12) = 1/36$, $P(11) = 2/36$ and $P(10) = 3/36$, calculated by applying the law of addition.

A full table of the possible scores and their associated probabilities can now be made.

Score	12	11	10	9	8	7	6	5	4	3	2
Probability	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

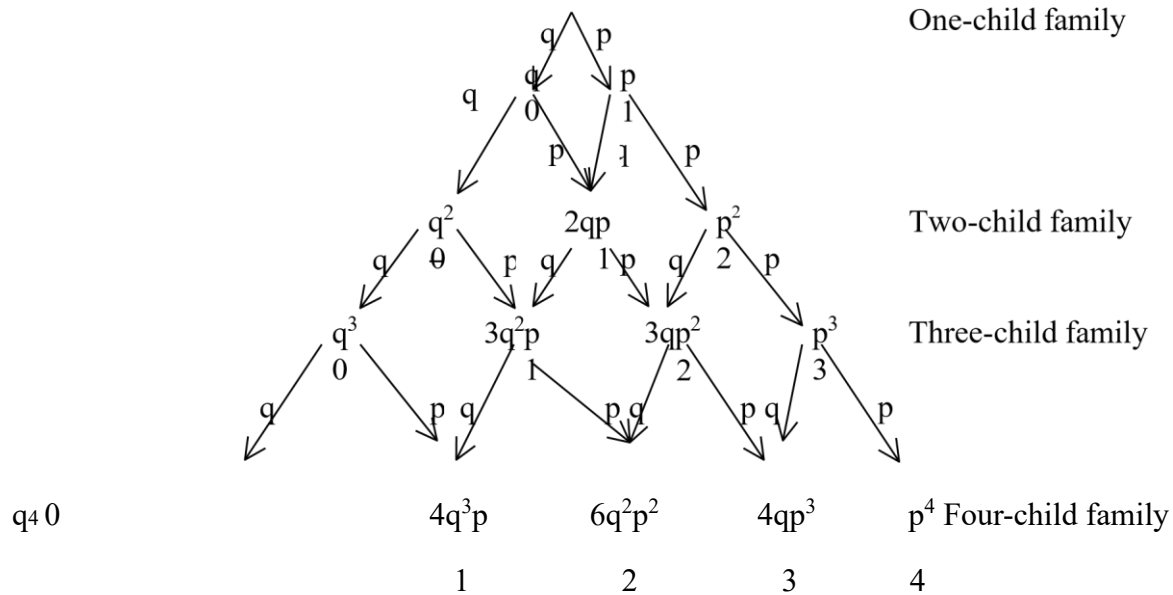
A distribution of all possible outcomes together with their associated probabilities such as the one shown above is known as a probability distribution. It will be seen that if all the probabilities are added together, they make 1, which means that it is certain that the score from throwing two dice will be a whole number between 2 and 12 inclusive. The probability distribution adds up to 1, because the distribution covers all possible outcomes. There are basically three types of probability distributions, i.e., Binomial, Poisson and Normal.

1.4.1 The Binomial Distribution

1.4.1.1 Using the Algebraic Method

One of the commonest probability distributions is associated with the repetition of events where there are only two possible outcomes. For instance, when a baby is born, it is either

a boy or a girl. The probabilities are about equal (0.5 each). To make the example more general, however, we shall call the probability of a boy being born (p), and the probability of a girl being born (q). Then $q + p = 1$, as these are the only possible outcomes of a birth. This can be illustrated as follows:



The figure above considers the probabilities of having different numbers of boys in one, two, three, and four-child families each horizontally line of figures shows all the possibilities for a family with the stated number of children. The term in the upper part of each group of figures is the probability of having the number of boys shown in the lower part of the group.

Thus, in a three – child family, the probability of having two boys is $3qp^2$. The diagram suggests the reason. To get this group of figures require two “moves” (i.e. births, shown by lines connecting the groups) of probability p each, and one of probability q . The probability of doing this once is $q \times p \times p = qp^2$ (multiplication law). But there are three possible ways of doing this (ppq , qpq , qpp), i. e., there are three alternative routes to the group of figures in question (boy, boy, girl then boy, girl, boy, then girl, boy, boy) the total probability of reacting this group is therefore $qp^2 + qp^2 + qp^2 = 3qp^2$ (addition law)

It will be observed that the terms in the upper part of each line of group of figures are the terms of a familiar algebraic Expression:

Note:

$ \begin{aligned} & q + p \\ & q^2 + 2qp + p^2 \end{aligned} $	$ \begin{aligned} & = (q+p)^1 \\ & = (q+p)^2 \end{aligned} $
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$$\begin{array}{l}
 q^3 + 3q^2 p + 3qp^2 + p^3 = (q + p)^3 \\
 q^4 + 4q^3 p + 6q^2 p^2 + 4qp^3 + p^4 = (q + p)^4
 \end{array}$$

From the above algebraic equations a Pascal triangle of the coefficients can be formed as follows:

$$\begin{array}{cc}
 1 & 1
 \end{array}$$

				1	2	1			
			1	3	3	1			
		1	4	6	4	1			
	1	5	10	10	5	1			
	1	6	15	20	15	6	1		
	1	7	21	35	35	21	7	1	
1	8	28	56	70	56	28	8	1	

The above Pascal Triangle can be used to expand any given algebraic expression of the form $(q + p)^n$. In fact it can be shown that the probabilities associated with all possible outcomes of births (or any other kind of event with only two possible outcomes) are given by the terms in the expansion of $(q + p)^n$. This is known as a “**binomial distribution**”. Expanding the equation using the Pascal triangle shown above can solve the binomial distribution.

Example

What is the probability of there being no boys, 1 boy, 2, 3, 4, 5, or 6 boys in a family of six children?. The answer is given by the terms of the expansion of $(q + p)^6$ which are $q^6 + 6q^5p + 15q^4p^2 + 20q^3p^3 + 15q^2p^4 + 6qp^5 + p^6$, but if $p = q = 0.5$, this can be restated as follows.

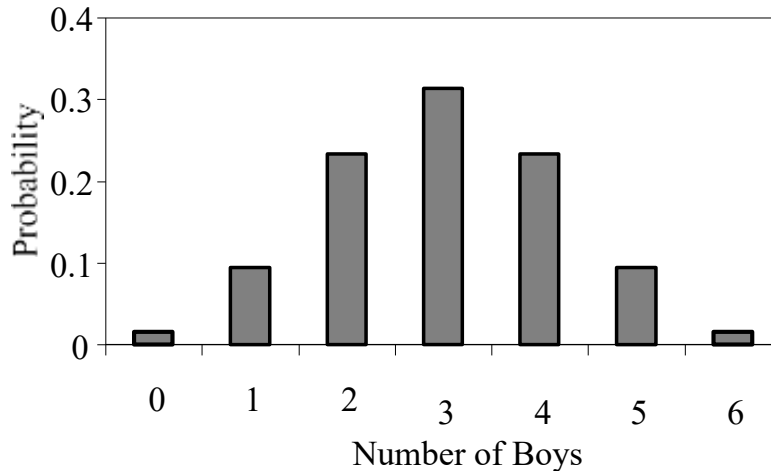
$$(0.5)^6 + 6(0.5)^6 + 15(0.5)^6 + 20(0.5)^6 + 15(0.5)^6 + 6(0.5)^6 + (0.5)^6$$

0 boy 1boy 2boys 3boy 4boys 5boys 6boys since $(0.5)^6$
 $= 1/2^6 = 1/64$, these probabilities can be written as

$$1/64, 6/64, 15/64, 20/64, 15/64, 6/64, 1/64 \text{ or}$$

$$0.016, 0.094, 0.234, 0.313, 0.234, 0.094, 0.016$$

These results can be shown by histogram as follows:



It can be noticed that from the figure above that:

- I. The distribution is symmetrical – there is just as much likelihood of having six boys (and no girls) as there is of having no boys and (six girls). This is because $p = q = 0.5$. Only when this is so is a binomial distribution symmetrical.
- II. The mean is three, that is, if one divided the number of boys in six –child families by the total number of such families, one would expect an answer of three.
- III. The greatest probabilities clusters around the mean, that is, most six – child families will have a bout half boys and half girls.
- IV. The greater the deviation from the mean, the smaller the probability of that number occurring. It is very unlikely that all six children will be boys, or all six girls.

This is of no particular help to parents with five children, all girls, who are wondering whether to try once more for a boy. The probability that the next child will be a boy is still about 0.5 (unless there is some genetic bias to produce one sex rather than the others). But assuming there is no genetic bias, it does mean that for every 1000 six – child families about 16 will have six girls and no boys, 94 will have five girls and a boy, 234 will have four girls and two boys, 313 will have three girls and three boys, 234 will have two girls and four boys, 94 will have one girl and five boys, and 16 will have six boys and no girl. If genetic bias were suspected, a binomial distribution would provide the norm against which to compare frequencies from a random sample of families.

1.4.1.2 Using Combination Method

Problem

Imagine we have an urn containing 100,000 marbles, 40,00 of which are red. A random sample of 4 marbles is taken, what are the probabilities that in this sample there are 0 red marbles, 1 red marble, 2 red marble, 3 red marbles and 4 red marbles respectively. (i.e. what probabilities will the random variable value 0, 1, 2, 3, and 4 take)?

Solution

In this problem we first note that p , the constant probability of a success is $40,000/100,000 = 0.4$ so q the probability of failure is $1 - 0.4 = 0.6$

So if x is the number of reds we can say that the probability of x is in some way is related to $p^x q^{n-x}$ (note p^0 and q^0 are both equal to 1)

The number by which we must multiply $p^x q^{n-x}$ is the number of combinations of n items n

13 taking x at a time i.e. $() x$

n
 $()$ is given by $n/(n-x)!x! x$

Note

(a) number of combinations of 5 items taking 3 at a time is calculated as:

$${}^5C_3 = \frac{5!}{(5-3)!3!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{(2 \times 1) \times (3 \times 2 \times 1)} = 10$$

(b) number of combinations of 15 items taking 5 at a time is calculated as:

$${}^{15}C_5 = \frac{15!}{(15-5)!5!} = \frac{15 \times 14 \times 13 \times 12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{(10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1) \times (5 \times 4 \times 3 \times 2 \times 1)}$$

$$= \frac{15 \times 14 \times 13 \times 12 \times 11}{5 \times 4 \times 3 \times 2 \times 1}$$

A shorter way of writing out this last calculation is cancel out factorials.

$$\text{Thus: } \frac{15!}{(15-5)!5!} = \frac{15 \times 14 \times 13 \times 12 \times 11 \times 10!}{10! \times 5!}$$

$$\text{The } 10! \text{ Cancel leaving } = \frac{15 \times 14 \times 13 \times 12 \times 11}{5 \times 4 \times 3 \times 2 \times 1}$$

Where we have only two possible different characteristics in a population then in a sample of n items the probability of 0, 1, 2, 3 ----- i.e. x items having a given characteristic are given by the formula.

$$P(x) = {}^nC_x p^x q^{n-x}$$

Applying this formula for all the random variable values between $x = 0$ and $x = n$ gives the Binomial Distribution.

and

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15

Example

Using our example where $p = 0.4$ and $q = 0.6$ and $n = 4$

$$P(x) = \binom{4}{x} 0.4^x \times 0.6^{4-x}$$

$$\text{Since } \binom{4}{x} = \frac{4!}{(4-x)!x!} = \frac{4 \times 3 \times 2 \times 1}{(4-x)!x!} = \frac{24}{(4-x)!x!}$$

then

$$P(x) = \frac{24}{(4-x)!x!} \times 0.4^x \times 0.6^{4-x}$$

Therefore Binomial distribution layout is

X (No of red marbles)	P(x) (ie probability of x marbles in sample)
0	$\frac{4!}{(4 \times 3 \times 2 \times 1) \times 0!} \times 0.4^0 \times 0.6^{4-0} = 1 \times 1 \times 0.6^4 = 0.1296$
1	$\frac{24}{(3 \times 2 \times 1) \times 1!} \times 0.4^1 \times 0.6^{4-1} = 4 \times 0.4 \times 0.6^3 = 0.3456$
2	$\frac{24}{(2 \times 1)(2 \times 1)} \times 0.4^2 \times 0.6^{4-2} = 6 \times 0.4^2 \times 0.6^2 = 0.3456$
3	$\frac{24}{1! \times (3 \times 2 \times 1)} \times 0.4^3 \times 0.6^{4-3} = 4 \times 0.4^3 \times 0.6^1 = 0.1536$
4	$\frac{24}{0! \times (4 \times 3 \times 2 \times 1)} \times 0.4^4 \times 0.6^{4-4} = 1 \times 0.4^4 \times 0.6^0 = 0.0256$
	<u>$= 1.0000$</u>

$$0! = 1 \text{ and } n^0 = 1$$

Activity

You have an urn in which there are 100 balls, 30 of the balls are blue. You take a random sample of 6, what are the probabilities that in your sample there are 0, 1 2 3 4 5 and 6 blue balls respectively.

1.5 Poisson Distribution

Problem

Imagine that in our 100,000 marble urn there are now only 120 red marbles but that our sample size is to be 1,000. What are the probabilities of having 0, 1, 2, 3, 4 etc red marbles in our random sample?

Solution

In this instance p is obviously only $120/100,000 = 0.0012$ and so employing the Binomial distribution will involve using the formula.

$$P(x) = \frac{1000!}{x! (1000 - x)!} \times 0.0012^x \times 0.9988^{1000 - x}$$

where x takes every value from 0 to 1,000.

In this situation computing only ($x=0$) means that we will have to find 0.9988^{1000} , while computing $P(x = 2)$ requires us to find $1000!/998! \times 0.0012^2 \times 0.9988^{998}$. Clearly, a shortened method would be necessary in this situation.

The Poisson distribution provides such a short cut and gives us a very good approximation to the Binomial distribution. The formula for this distribution is

$$P(x) = e^{-a} \times a^x / x!$$

Where

a = average number of items per sample given by formula $a = np$

e is a constant (rather like π) which means that a table for e^{-a} can be prepared so that finding e^{-a} for any value of " a " simply involves working up the table, which is given in statistical text books (Appendix 1).

Example

Let's take our current illustration

Here $P = 0.0012$ and $n = 1,000$ and so $a = 0.0012 \times 1000 = 1.2$. The table shows that $e^{-1.2}$ (e^{-a}) = 0.3012.

Applying the Poisson distribution formula:

$$P(x) = 0.3012 \times 1.2^x / x!$$

and

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17

Since we want the probability of having 0, 1, 2, 3, 4, etc red marbles in one sample of 1,000 the layout we need is as follows.

X	P(x)
No red marbles	(i.e. probability of x red marbles in the sample)
0	$0.3012 \times 1.2/0! = 0.3012.$
1	$0.3012 \times 1.2 /1! = 0.3614$
2	$0.3012 \times 1.2/2! = 0.2169$
3	$0.3012 \times 1.2/3 = 0.0867$
4	$0.3012 \times 1.2/4! = 0.0260$
5	$0.3012 \times 1.2/4! = 0.062.$

5 since we know that the whole probability must add up to 1 and these p(x)s must collectively be the balance required to make the total equal 1 □

Activity

In Nairobi City, 10 per cent of the residential houses lack bathroom.

(a) Find the probabilities that a random sample of ten houses will contain 0, 1, 2, 3, 4 , 6, or more houses without bathrooms using

(i) the Binomial distribution

(ii) the Poisson distribution

(b) Draw a grouped histogram of the two distributions.

(c) From the Binomial distribution find the probability that the sample will contain less than two bathroom houses. Work to 4 decimal places.

1.6 The Normal Probability Distribution

The normal probability distribution is a hypothetical distribution to which a large number of actual distributions approximate.

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18

Illustration

The table below sets out the annual rainfall figures (inches) correct to one decimal place for Nakuru for 50 years. I e 1917 to 1966 inclusive.

Annual rainfall (inches) at Nakuru 1917 – 1966 inclusive

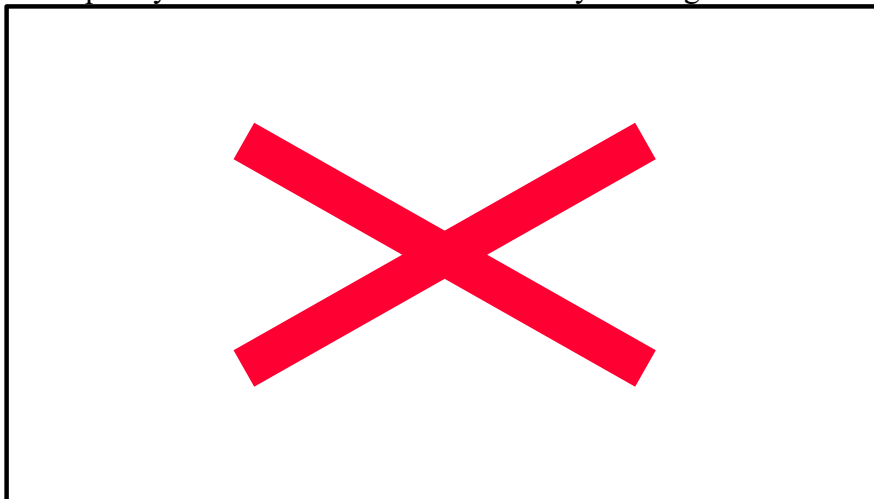
19 17 - 19	1920 - 29	1930 - 39	1940 - 49	1950 - 59	1960 - 66
	27.5	37.0	24.1	24.1	37.0
	19.6	30.1	24.8	27.1	22.4
	27.6	28.8	19.2	23.5	19.7
	25.7	21.5	20.2	21.4	22.4
	28.6	20.7	24.0*	28.4	19.8
	30.3	27.6	21.7	21.2	31.8
	23.7	24.1	27.5	24.5	33.6
24.2	30.9	23.3	21.3	25.8	
25.8	28.8	21.6	24.9	29.3	
27.9	24.2	24.7	22.2	17.8	

* Is 23.98 and therefore counted in-group 22 – 24 below.

The above figures can be put into the form of a grouped frequency distribution as follows:

Annual Rainfall	16-18	18-20	20-22	22-4	24-6	26-8	28-30	30-2	32-4	34-6	36-8
Frequency	1	4	8	7	12	6	5	4	1	0	2

This frequency distribution can be illustrated by a histogram as shown below.



and

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19

The mean and standard deviation have been calculated to be:

Mean (m) = 25.3 inches

Std. Deviation (s) = 4.3 inches

As this is an actual distribution, we are not surprised that it does not possess the regularity of the binomial distribution shown above. The question which arises is if there is a regular hypothetical frequency distribution i.e. a probability distribution to which this would approximate in the long run (i.e. with a much longer run of years, assuming no longer term climate change)

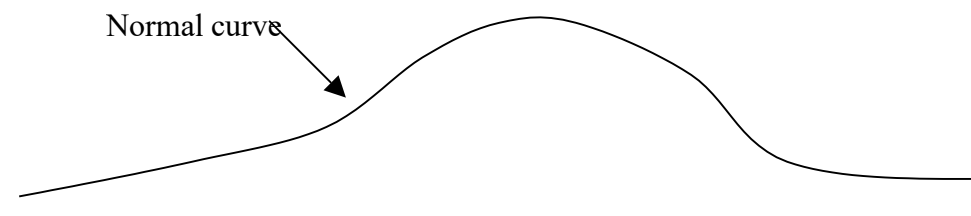
Because if there is, and if its mathematical properties can be determined, it will provide a means of calculating rainfall probabilities for Derby.

There are both empirical and logical reasons for believing that there is.

Empirically, the shape of the histogram shown above, while not perfectly regular is by no means random and shows some symmetry. Most of the values are clustered near the mean, and frequencies decline about equally in both directions with increasing distance from the mean. Rainfall data for other stations with widely differing regimes have similar characteristics (except for stations with very low rainfalls). Logically one would expect such a pattern.

Rainfall appears to be affected by two groups of factors: one dominant group that act consistently and causes clustering about the mean (e.g. latitude, altitude and exposure to sea influence) and another group of factors that acts randomly and causes in the long run a symmetrical variation about the mean (e.g. the pressure pattern).

It is generally agreed, therefore, that a frequency distribution of annual rainfall totals such as that just presented has a corresponding probability distribution, which is symmetrical with the highest probabilities near the mean, and with probability decreasing as distance from the mean increases. It has a general bell – like shape as indicated below, where it is shown by a curve in relation to the rainfall data discussed above. It is called a “normal distribution” and the curve is called “normal curve.”



It can be said that whenever there appears to be two groups of factors affecting a variable one dominant group acting consistently and another acting randomly, then there are a priori grounds for expecting that a frequency distribution of values will in the long run,

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20

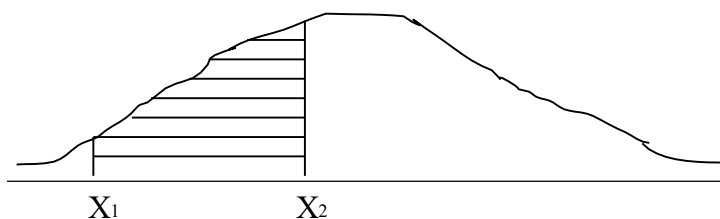
approximate to a normal distribution. For instance the roundness of pebbles on a beach, temperature and other weather data, crop yields under uniform conditions, pedestrians and traffic densities for particular places at particular times of the week. Also population data for towns such as average age, percentage employed in distributive trades, or rates of population growth, are a few variables that are likely to show a tendency towards a normal distribution.

1.7 Properties of a Normal Curve

In both binomial and normal distributions, probabilities fall away symmetrically from peak at the mean. But, whereas the normal distributions represented by a curve, the binomial is represented by a histogram. This is because the normal distribution is related to values on a continuous scale. Rainfall can have any value whereas a number of boys in a family must be a whole number. For example one can have 4.75 inches of rain, but not 4.75 boys. Similarly a normal curve gives probabilities for any values of rainfall, while Binomial can only do so for discrete values such as whole numbers.

Implicit in the normal curve being related to a continuous scale is the fact that it is related to an infinite number of possible values such as 25, 25.1, 25.01, and 25.001. The probability of any particular value occurring is therefore infinitesimally small and can be regarded as zero. One can only speak of the probability of values occurring within a

given range and this is shown by the area under the appropriate part of the curve, and not by its height as shown.



In the figure above the probability of values lying between x_1 and x_2 is proportional to the shaded area under the curve. Furthermore since the normal curve describes a probability distribution, the total area under the curve is proportional to the total probability, which equal to one (1).

As the normal curve is symmetrical about the mean half the area beneath the curve lies on each side of the mean. This implies that in a frequency distribution which approximate to normal, 50% of the values will be less than the mean, and 50% more than the mean, in

and

Cartography

21

other words, the probability of a value being less than the mean is equal to the probability of it being greater than the mean and both are equal to 0.5.

Less obvious property of the normal curve arises from the mathematical equation determining its shape and is related to its standard deviation () and the mean (). It is found that approximately 68% of the area under a normal curve lies within one standard deviation of the mean, 95% of the area lies within two standard deviation of the mean and 99.7% of the area lies within three standard deviation of the mean.

With this information, it is possible to estimate certain probabilities about any variable, whose distribution is approximately normal, if we know the mean and standard deviation. For instance, we have seen that the distribution of annual rainfall totals is approximately normal, and that on the basis of data already quoted, the mean annual rainfall of Derby is 25.3 inches, with a standard deviation of 4.3 inches. We can therefore make the following probability statements about Derby's annual rainfall (R)

Note

- (i) The probability that R will be less than 25.3 = 0.5 i.e. $P(R < 25.3) = 0.5$ equally.
- (ii) The probability that R lies between 21 inches and 29.6 inches is 0.68 (21 inches is 1 std deviation below the mean $25.3 - 4.3 = 21$), and 29.6 in is 1 SD above the mean (i.e. $25.3 + 4.3 = 29.6$). Similarly,
- (iii) $P(12.4 < R < 33.9) = 0.95$ (i.e. two SD's below and above the mean)
- (iv) $P(12.4 < R < 38.2) = 0.997$ (i.e. three SD's below and above the mean)

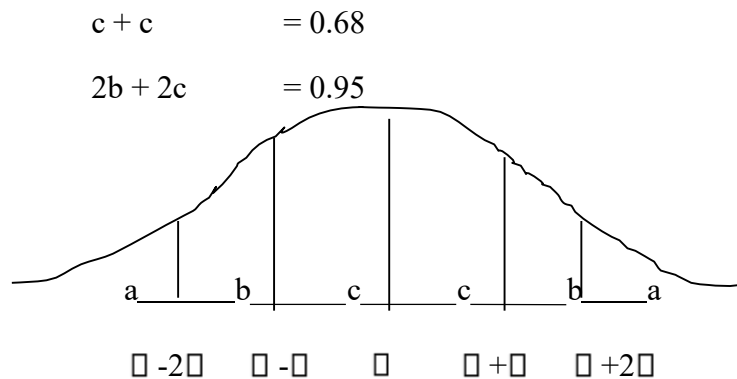
From these statements, we can deduce the number of occasions in 50 years when we would expect the annual rainfall to be within the stated ranges, and it is interesting to compare these expected frequencies with the actual frequencies between 1919 and 1966.

		Expected	Actual
R less than 25.3 inches	0.5×50	25	29
R more than 25.3 inches	0.5×50	25	21
R between 21 and 29.6 inches	0.68×50	34	36
R between 16.7 and 33.9 inches	0.95×50	47 or 48	48
R between 12.4 and 38.2 inches	0.997×50	50	60

The discrepancies between the two sets of figures could simply be due to the expected frequencies being derived from a regular probability distribution (i.e. the normal distribution), while the actual frequencies contain the usual incidental irregularities.

Activity

The figure below represents the probability of any normally distributed variable lying between the values indicated. Thus $a + b + c = 0.5$



(1) What are the values of a , b , and c ?

(2) What is the probability of x being less than:

- (i) $\square - 2\square + 2$ (ii) $\square - \square$ (iii) $\square$ (iv) $\square + \square$ (v) $\square$

(3) What is the probability of x being more than:

- +2 (v) $\square$
- (i) $\square - 2\square$ (ii) $\square - \square$ (iii) $\square$ (iv) $\square + \square$ (v) $\square$

(4) The 'A' horizon of a podzolic soil in a locality of generally uniform terrain has a mean depth of 10 cm, with a standard deviation of 2 cm. Assuming the depth of the 'A' horizon to be normally distributed, what is the probability that a randomly selected soil profile will have an 'A' horizon of the following depths.

- (i) less than 8 cm
- (ii) more than 14 cm
- (iii) between 10 and 12 cm
- (iv) exactly 10 cm
- (v) between 10 and 14 cm?

and

Cartography

23

Suppose we wish to determine the probability of the annual rainfall being less than 20 inches. We know that the mean annual rainfall is 25.3 inches and the standard deviation is 4.3 inches, so that 20 inches is not a whole number of standard deviations from the mean. It is 5.3 inches from the mean, which is more than one S D but less than two S D's from the mean. To solve this problem, we need to know probabilities corresponding to values between whole number standard deviations from the mean. To do this two steps are necessary.

- 1 Convert the value in question (i. E. 20 inches) into a "Z – score". This is simply the number of standard deviations of a value above or below the mean.

Thus, if the value in question is x, then

$$Z = (x - \text{mean}) / \text{standard deviation}$$

This may be written as either:

$$Z = x - \bar{x} / s$$

Where $\bar{x}$ and s are the mean and standard deviation of a population. Or

$Z = x - \bar{x} / s$, where $\bar{x}$ and s are the mean and standard deviation of a sample.

Note

In our problem:

X = 20 inches. Mean = 25.3 in

Standard deviation = 4.3 in.

That is

$$Z = (20 - 25.3) / 4.3 = -1.2$$

- 2 Consult one of the tables of probabilities associated with different Z scores in a normal distribution (Appendix 2).

It should be noted that P value in the table relate to only half of a normal distribution and that one half of a normal distribution is a mirror image of the other. Therefore a negative Z score has the same P value as a positive Z score of equal magnitude, that we can ignore the sign of Z when using the normal distribution tables associated with the Z score.

and

Cartography

24

In our problem Z is 1.2 and we must know the probability of a variate being more extreme than, i.e. (lying beyond) this Z score. The table value of P corresponding to $Z = 1.2$ is 0.115. It follows that the probability of the rainfall being less than 20 inches is 0.11 or 0.12 meaning that an annual rainfall of under 20 inches can be expected at Derby eleven years in every hundred years.

Activity

What is the probability that rainfall at Nakuru will be at least 28 inches.
What is the probability that it will be between 20 inches and 30 inches?

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LESSON 2: SAMPLES AND ESTIMATES.

2.1 Introduction

Geographers are often required to undertake research in which data is collected, analyzed and presented. To do this effectively, knowledge of sampling techniques is important. In this lesson the basic concepts and procedures of sampling will be discussed. A distinction will be made between population and samples and between population parameters and sample statistics. Finally a discussion of the sampling distribution and the standard error and how it is used to estimate population parameters will be done.

2.2 Objectives

and

Cartography

25

By the end of this lesson the learner should be able to:

- Distinguish between population and samples, and also between population parameters and sample statistics
- Explain and use the various sampling procedures to collect data from a given population
- Estimate population parameters from sample measurements and counts and also estimate required sample size from a given population.

2.2 Population and Samples

In statistics the word population has a technical meaning, which may have nothing to do with people. Statistically population may be defined as a complete set of counts or measurements derived from all objects possessing one or more common characteristics.

A population may have a finite number of individual counts or measurements (e.g. the size of factories in a town); it may be almost infinite (e.g. the composition of pebbles on a beach); or it may be infinite (e.g. the slope of the land at point locations on a map), there being an infinite number of points on a map.

It is often neither necessary nor practicable to obtain all the counts or measurements for a complete population. One can obtain fairly good, and for most purposes perfectly adequate, approximations by measuring or counting only a fraction of all the population, and thereby save a great deal of time and trouble. The data so collected would constitute a “sample” which may be defined simply as part of a population.

A sample is therefore, used as a substitute for populations, to save time or where measuring or counting the whole population is impractical. If it is to be a reliable substitute, it must be a “representative sample”, that is, it must represent the variability in the population as closely as possible. To achieve this, it is important to avoid “bias” which results in the inclusion in the sample of unrepresentative proportions of data from different parts of the population.

If samples are selected entirely by personal judgement on the part of the sampler, “bias” is almost bound to occur. Subjective selection cannot free itself from subconscious predilections of the sampler for picking up small pebbles rather than large, observing plants with striking colours more often than those of a more subdued colours, or interviewing attractive members of the opposite sex. Even if subjective selection is made without bias, this cannot be demonstrated, and there can be little confidence that the sample is both to be and to be seen to be without bias. The selection of individuals should therefore not depend upon the judgement of the sampler. Methodical procedures must be used, which reduce the subjective element in selection to a minimum.

and

Cartography

26

2.3 The Sampling Frame

Sampling Frame is a statistical population from which to take the sample. Often the true population is not amenable to sampling in the time available and a substitute has to be found. For instance if one wants to sample 10% of a population of Kenya, it will be impossible to question 10% of all individual people. Instead 10% of households can be sampled. All the people of Kenya constitute the true population, but it is the households of Kenya, which constitute the sampling frame from which a 10% sample is taken.

The sampling frame may be a serious source of bias, depending on how representative it is of the true population being studied. For example weekday shoppers would provide a poor sampling frame from which to draw a sample of public opinion. It would contain too many housewives and unemployed and not enough people in employment.

Sampling frames may be either spatial or non-spatial. A "spatial sampling frame" is one in which location is an essential part of the variability of individual observations, and where it is therefore important that samples are representative of the full range of locational variability. If, for example, one is sampling a map to determine the amount of woodland in the map area, one must ensure that all parts of the map are duly represented.

This may be done in one of three ways:

- (i) Point - sampling e.g. by sample grid-intersections.
- (ii) Line - sampling e.g. by a sample of linear transects or cross- sections.
- (iii) Area - sampling e.g. by a sample of quadrants or grid - squares.

The sampling frames in these cases will be the populations of all possible gridintersections, linear transects, and quadrants on the map respectively.

Although geographers are particularly interested in spatial sampling frames, they also make frequent use of the non-spatial variety, where the in locational variability of individual observations is not considered. The sampling frame is then simply the total collection of phenomena under consideration, such as all the households in the parish, all the shops in the Central Business District (C B D), or all the shoppers in the main street. It is still important to sample the full range of variability within the population, but this can be done without reference to their location. For example, by using an electoral register (census register), or telephone directory or by simply stopping people outside a supermarket.

Whether one uses a spatial or non-spatial sampling frame depends upon the objectives of the inquiry. If it is to find out something non-spatial about an area such as a whole, (such as the number of bookshops in a city's C B D - assuming that C B D is already defined), then a non-spatial sampling frame may be adequate. If it is to find out something spatial about an area as a whole (such as the amount of land occupied by bookshops in the C B D), then a spatial sampling frame is required.

Activity

Design two any research activities where you will apply in one spatial and in another non-spatial sampling frames

2.4 Sampling Procedures.

Having decided upon a suitable sampling frame, it is then necessary to select individuals for the sample, by a method which is as mechanical as possible.

The two most common sampling methods are:

- (i) Random sampling
- (ii) Systematic sampling.

2.4.1 Random Sampling

A random sample is defined as one in which any one individual measurement, or count in the population is as likely to be included as any other. The usual way of obtaining a random sample is by the use of a table of random numbers.

Example

Suppose we wish to obtain a random sample of 20 farms from a given area on a map, with a view to investigating various aspects of the farm economy of the area. How we will go about it?

- (1) We begin by choosing a sampling frame, which will ensure a random spacing of farms. Using the intersections or squares of a superimposed grid often does this. Let

the sampling frame in this case be all the farms with their farmhouse nearest the southwest corner of the grid square in which they are located". This means that we first require a random sample of grid squares, so that we can select the farm nearest the southwest corner in each case. Because some of the squares may contain no farms, we shall obtain a random sample of 25 squares, and then choose the first 20 that contain farmhouses.

- (2) To obtain the sample of grid squares the table of random numbers is used (Appendix 3). Select a pair of digits from anywhere in the table, and work methodically across the table in any direction, reading off two digits at a time. Start at the bottom right hand

and

Cartography

28

corner of the table, and work up the right-hand column, to the top up then down the next column, up the next again, and so on.

- (3) The numbers are first used to determine the eastings of the sample grid squares. These can only run from 00 to 14, so we shall ignore most of the pairs of random numbers we read off. The first 25 pairs of digits that come within our range of eastings are set out in the left-hand column below.

For example the following are 25 randomly selected grid reference numbers from table of random numbers:

11	36
01	27
07	39
09	41
11	31
10	30
14	28
08	37
12	42
08	28
08	26
01	29
03	40
04	34
08	33
06	30
14	33
03	33
08	35
07	30
07	32
03	36
01	37
06	26
00	30

- (4) When sufficient eastings have been obtained, the reading off of random numbers is continued to obtain the same number of northings within the grid range of 25 to 42 (right-hand column) of table above.

- (5) We now have 25 four-figure grid references, beginning with 1136, these determine the grid squares in which our sample farms are located. From the 25 we select a sample of

and

Cartography

29

20 farms. In the case of those squares containing two or more farmhouses, we select only the nearest farmhouse to the southwest corner of the square.

- (6) It may be necessary to introduce additional rules to avoid having to make judgements about the selection of individual cases. For instance, if farm A lie on a line between two squares how do we select? For such cases, a rule is made to assign the farm to the square to the north and/or east of the line.

It will be seen that the rules are quite arbitrary. This does not matter, so long as they are simple, do not create bias, and are applied consistently so that the subjective element in the selection of individual cases is eliminated

2.4.1.1 Systematic Sampling

A systematic sample is one in which selection is made from the population at regular intervals. For instance, every 1/10th name on an electoral list, every 1/5th shop listed in a classified telephone directory, every alternate grid intersection; all these will yield systematic samples. It is generally quicker and simpler to obtain a systematic sample than a random sample, for random numbers are not required. A systematic sample also gives a more uniform cover of the population, for random samples tend to include clusters and leave gaps unless the random sample is very large. Thus, if one were estimating the proportion of woodland in a map area, and if the major area of woodland happened to be clustered in one locality, the estimate would be above the true value (other things being equal).

On the other hand, if there are regularities in the background population, a systematic sample may be biased. If one were to impose a rectangular grid over many American cities, a systematic sample based on the grid intersections could consist of data derived mainly from street corner locations.

Generally, a systematic sample is likely to be biased unless the background population from which it is taken is itself randomly distributed and many background populations are not randomly distributed. On the other hand, the bias in many systematic samples is small, and may be a price worthy paying for the time saved.

A compromise between random and systematic methods is sometimes used to simplify selection at a minimum cost in terms of bias. For example if one wanted to enumerate 10% of all households in Kenya one can perform a systematic random sampling as follows: Each enumeration district can be allocated a randomly selected digit, which can be called n . The enumerator is then instructed to call only at the n th, $(10 + n)$ th, $(20 +$

$n)$ th, etc, households as he walks systematically round his district. He is therefore spared the task of selecting his own sample randomly, while varying the digit from district to district randomly minimizes the bias of complete systematic sampling.

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Cartography

30

2.4.1.2 Stratified Sampling

A refinement of sampling procedure is the use of stratified samples, in which the population is divided into sub-sets, and separate samples drawn from each other. This is important if the population is made up of different size or character. Different procedures may be required for the different sub-populations.

Example of a problem where stratified sampling could be applied involves a comparison of two variables when their variability is not distributed in equal proportions over the area being studied. For instance, if one were investigating the relationship between altitude and vegetation on the limestone areas of a given region, it would be desirable to have adequate representation in each altitude band. But a simple random sample of points would be unlikely to provide this, because there is much more of the region in some altitude bands than in others. It could, however, be achieved by first stratifying the area into altitude bands of say 0-300m, 300-600m, 600-900m, etc. And then accepting the first 20 points in each altitude band that are selected randomly. When a population is stratified into sub-populations and then random selection of sub-samples done the samples are called “stratified random samples”. If the selection of sub-samples is made systematically then the results is called a “stratified systematic sample”.

2.4.1.3 Cluster Sampling

If the population is very widely dispersed, straightforward random or systematic sampling will involve a great deal of travel. Opinion polls in Britain, for example, generally use samples of about 200 people, which is about one in every 25,000 of the population. It saves an immense amount of time to sample from carefully selected clusters, e.g. from ten inner urban areas, ten suburban areas, and ten rural areas of varying location, social structure etc. A double sampling procedure is, therefore, involved first, selecting representative clusters (which is probably best done subjectively) and then selecting a random, systematic or stratified sample within each cluster.

A classification of formal Sampling Designs.

Type	Spatial			Non-Spatial
Procedure	Point	Line	Area	
Random				
Systematic				
Stratified				
Cluster				

Exercise

1. Explain why a geographer should have knowledge of sampling techniques

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Cartography

31

2. Describe each of the following sampling procedures and give two examples of a geographical research undertaking where each can be applied

 - (a) Random sampling
 - (b) Systematic sampling
 - (c) Stratified sampling
 - (d) Cluster sampling

2.4.1.4 Conclusion

Although sampling procedures are designed to minimize subjective selection, the choice of sampling frame (i.e. population or sub-populations from which the sample is selected) and the sampling method (i.e. method of selection from the sampling frame) does require common sense. There is no set of rules that will cover every sampling problems, and an investigator must be prepared to use his/her ingenuity in devising procedures that will provide reasonably unbiased data from a reasonable expenditure of time and effort.

2.5 The Use of Surrogates

Sampling is used to ease problems of data collection. Another commonly used device to achieve similar end is the use of surrogates, i.e. substitute variables that tell approximately the same story as the variables we are really interested in (the target variables). For example, unemployment rates are surrogates for economic status, infant mortality for social well being, newspaper circulation for literacy, distance for cost or time separating two locations etc.

In many ways surrogates are analogous to sampling because;

- a) They facilitate data gathering and in some circumstances are the only ways of obtaining any data at all. For instance finding data on a variable such as illiteracy in many developing countries, or data on a target variable, which is too diffuse to be pinned down to direct measurement such as social well being or economic status).
- b) Surrogates are subject to random “error” (i.e. variations in the surrogate will differ randomly from variations in target variable), which may prevent an investigator achieving conclusive results, since the variations he is investigating are observed by random variations in the surrogate. But, whereas the level of random error in samples can be estimated from probability theory, the same is not true of surrogates. Only if we have some target variable data with which to compare the surrogate data can we estimate the amount of random error in the latter. The higher the correlation between

the two variables, the less random error there will be, and the more reliance can be placed on one variable as a surrogate for the other.

and

Cartography

32

- c) Surrogates are also subject to bias (i.e. variations in the surrogate may differ systematically from variations in the target variable). For example use of occupations of heads of households to determine employment structure will overestimate the kind

of jobs undertaken by older men, and underestimate those undertaken by the young and by women. On the other hand, whereas in sampling there are standard procedures to minimize bias, the same is not true for the selection of surrogates. One simply has to rely on the application of common sense and geographical training and knowledge of the factors relating a surrogate to its target variable.

2.5.1 Population Parameters and Sample Statistics

Samples are used to obtain estimates of numerical characteristics of populations where it is impractical to measure or count the whole population. It is conventional to use the term “population parameter” to denote a numerical characteristics of a population and “sample statistics” to denote a numerical characteristic of a sample. A sample statistics such as mean roundness of 100 randomly selected pebbles, generally provides the only estimate available of the corresponding population parameter such as the mean roundness of all the pebbles on a beach. The most widely used parameters and statistics are the mean and standard deviation, together with the number of variates from which they are derived. Following conventional notation indicates them:

	N	n	Population Parameter, Sample Number of variates
Mean			$\bar{x}$
Standard deviation			s

2.6 Sampling Distribution and the Standard Error

It is highly unlikely that any one sample, no matter how unbiased, will yield values of $\bar{x}$ and s, which give precise estimates of the corresponding population parameters μ and σ . It is equally unlikely that two different samples drawn from the same background

population, will have exactly the same values of $\bar{x}$ and s. Indeed, if one took a large number of samples from the same population, it is probable that each would produce a different mean and standard deviation. A distribution of such sample statistics (e.g. of the means of samples drawn from a common background population) is known as a “sampling distribution.” If the samples are all unbiased, it is reasonable to expect that their means will be distributed symmetrically about the corresponding population mean, that is, some

and

Cartography

33

sample means will be bigger than the population mean, and a roughly equal number will be smaller. The same is likely to be true of other sample statistics in relation to their corresponding population parameters.

Now in practice we do not take a number of samples but only one and we do not know the population parameters but can only estimate them from our one sample. Nevertheless, by hypothesizing sampling distributions based upon a large number of imaginary samples, mathematicians have been able to provide us with a measure of the accuracy of parameter estimates based on a single sample. This can be done based on the central limit theorem and the concept of the standard error.

The “ Central Limit Theorem” states that if we imagine taking all possible samples of similar size from a single population, the sampling distribution of their means will be approximately normally distributed about the population mean, whatever the character of the population distribution, provided the sample are fairly large (say over 30). Also, the standard deviation of the sampling distribution of means is given by $(\sigma/\sqrt{n})$, i.e. standard deviation of population divided by square root of size sample.

The standard deviation of a sampling distribution is called **Standard Error of the sample statistic**. Therefore the standard deviation of the sampling distribution of means

is called the standard error of the mean ($SE_{\bar{x}}$). Given as.

$$(SE_{\bar{x}}) = \frac{\sigma}{\sqrt{n}}$$

σ = Std deviation of population

n = size of sample

The standard error is the key to accurate estimation of population parameters from sample statistics

2.7 Estimating From Sample Measurements

Suppose we wish to estimate the mean roundness of pebbles on a stretch of beach.

We take a random sample of 100 and find that $\bar{x} = 50$ and $s = 10$.

The value of $\bar{x}$ gives us a rough estimate of μ , but how rough? What is the likelihood of the true value μ being as high as 60 or as low as 40?

and

Cartography

34

We begin by estimating the standard error of the sample mean using

$$\text{Formula } S E \bar{x} = \frac{s}{\sqrt{n}} \sqrt{\quad}$$

$$\text{Thus } S E \bar{x} = 10 / \sqrt{100}$$

We know that $n = 100$, but can only estimate s being roughly equal to $s (= 10)$. Thus

our estimate of $S E \bar{x}$ is $10 / \sqrt{100} = 1$

We can now imagine a sampling distribution of $\bar{x}$, centred on μ (which we are trying to estimate), which is normal (Central Limit Theorem) and has a standard deviation ($S E \bar{x}$) approximately equal to 1. We also have one value from this sampling distribution ($\bar{x} = 50$).

Because the distribution is normal, the following probability statements can be made.

- (i) The probability that a single value of x lies within 1 S E of μ is 0.68.
- (ii) Therefore the probability that 50 lies within 1 S E of μ is 0.68
- (iii) Similarly the probability that 50 lies within 2 S Es of μ is 0.95
- (iv) And the probability that 50 lies within 3 S Es of μ is 0.997 The above statements can be reserved as follows:

- (i) The probability that $\bar{x}$ lies within 1 S E of 50 = 0.68
- (ii) The probability that $\bar{x}$ lies within 2 S Es of 50 = 0.95 (iii) The probability that $\bar{x}$ lies within 3 S Es of 50 = 0.997

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And since $S E \bar{x} = 1$, we can restate this as:

- (i) There is a 68% probability that $\bar{x}$ lies between $50 - 1$ and $50 + 1$ i.e. between 49 and 51.
- (ii) There is a 95% probability that $\bar{x}$ lies between 48 and 52.

- (iii) There is a 99.7% probability that $\bar{x}$ lies between 47 and 53.

These levels of probability are usually called “confidence levels”, and the limits within which a parameter lies with a specified level of confidence are known as “confidence limits”. The range between confidence limits is called the “confidence interval”.

Therefore the 95 per cent confidence limits for pebble roundness on our beach are 48 and 52, and the 95% confidence interval is 4. We can say with 95 per cent confidence that the mean roundness for the whole population of beach pebbles lies between 48 and 52. A value outside this range is sufficiently improbable to be ignored for many purposes, and the 95% level of confidence is frequently regarded as giving tolerable limits for making population estimates

Note

The 68 percent confidence limits of $\bar{x}$ are: $\bar{x} \pm 1 S.E.$
 The 95 percent confidence limits of $\bar{x}$ are: $\bar{x} \pm 2 S.E.$
 The 99.7 percent confidence limits of $\bar{x}$ are: $\bar{x} \pm 3 S.E.$

We may however wish to ascertain limits for confidence levels other than 68% (which is too low for most purposes), 95% (the most commonly used confidence level), or 99.7. We must then use the table of z value probabilities associated with the normal distribution. Confidence levels express the probability of values (from a sampling distribution, which is normal) lying within specified limits on either side of the mean. The table of z – value probabilities gives the probability that values taken from a normal distribution will lie within z standard deviations on one side of the mean. Therefore the probability associated with a z-value in the table is half the corresponding confidence level. The procedure for using z-value tables to determine confidence limits is therefore as follows:

- (i) Halve the required confidence level and express it as a probability out of 1 (e.g. for a 90% confidence level, the probability required is 0.45.)
- (ii) Look up the corresponding Z-value in the table (for P = 0.45, Z = 1.6)
- (iii) The confidence limits are then $\bar{x} \pm Z S.E.$

In the case of the beach pebbles, where S E was about 1 and $\bar{x}$ was 50, the 90% confidence limits for mean roundness are 50 ± 1.6 i.e. 48.4 and 51.6

and

Cartography

36

It must be stressed that the above methods of determining confidence levels are only valid if the sample from which the estimates are made is fairly large (over about 30), and that they take no account of possible bias.

It is important to remember from the above discussion the following:

- (i) For a given size of sample, the standard error, and therefore the confidence interval, increases as the standard deviation of the sample increases, i.e. the greater the standard deviation of the sample, the less accurately can one estimate the population mean. This can be illustrated by considering how the 95% confidence limits for mean roundness of our beach pebbles would vary with different values for a sample standard deviation.

The relationship between standard deviation, standard error and confidence interval

n	$\bar{x}$	s	$SE = s/\sqrt{n}$	95% confidence limits	95% confidence interval
100	50	10	1	48 and 52	4
100	50	20	2	46 and 54	8
100	50	30	3	44 and 56	12
100	50	40	4	42 and 58	16

- (ii) For a given standard deviation of sample, the standard error and confidence interval decrease, and the confidence limits narrow as the sample size increases, i.e. the greater the sample size, the more accurately one can estimate the population mean. A table below can illustrate this.

$\bar{x}$	n	s	$SE = s/\sqrt{n}$	95% confidence limits	95% confidence interval
50	100	20	2	46 and 54	8
50	200	20	1.4	47.2 and 52.8	5.6
50	300	20	1.2	47.6 and 52.4	4.8
50	400	20	1.00	48.0 and 52.0	4.0

Exercise

In a survey among Kenyan villages, the diet of 100 randomly selected adult males was calculated to contain a mean daily calorific value of 2000, with a standard deviation of 250.

- (a) Calculate the 95% confidence limits for the mean daily calorific value of the diets of all adult males living in those villages.
 (b) Assuming that s remains 250 by how much would the sample have to be increased in order to have the confidence interval?

2.8 Estimates From Small Sample Measurements

The central limit theorem only applies to fairly large samples (over about 30). The larger the sample, the less the shape of a sampling distribution depends on the normality of the background population. Thus, while the sampling distribution of large sample means is used to estimate the population mean whatever the shape of the background population, the shape of the sampling distribution of small sample means depends upon the shape of the background population. If this is not known, there is no way of making a statistical estimate of the population mean from a small sample.

If we wish to estimate the mean of a normally distributed population from a small sample, two modifications to the procedure outlined above are necessary.

If we wish to estimate the mean of a normally distributed population from a small sample, two modifications to the procedure outlined above are necessary.

- i) It is found that a better estimate of the standard deviation of the population mean (σ) is obtained by multiplying the sample standard deviation (s) by $\sqrt{n/(n-1)}$. This is called the Bessel's Correction, and gives the "best estimate" of σ . Thus:

$$\sigma \approx s \sqrt{n/(n-1)}$$

It can be seen from the above equation that as n increases, $\{n/(n-1)\}$ approaches 1, and the difference between s and σ becomes slight. This is why the modification can be ignored when n is greater than 30.

- ii) The Z- table used in making estimates from samples needs to be replaced by what is called the t-table when the sample size is small. This is because the sampling distribution of small-sample means is not normal, but follows what is called a "Student's t-distribution. The shape of the t-distribution depends on the size of the sample, and as the latter increases, the t-distribution becomes more and more like the normal distribution.

Example

A static count of pedestrians outside a supermarket on nine Saturday mornings gave a mean flow of 2500 per hour, with a standard deviation of 400. What are the 95% confidence limits for the mean Saturday morning pedestrian flow?

Solution

Step 1: Find the best estimate of the standard deviation of the population from the formula:

$$\sigma \approx s \sqrt{n/(n-1)} = 400 \sqrt{9/8} = 424$$

and

Cartography

38

Step 2: Find the standard error of the sampling distribution from the formula:

$$SE = s / \sqrt{n} = 424 / 9 = 141$$

Step 3: Look up in the t-tables how many standard errors either side of the mean contain 95% of the t- distribution, when the size of the sample is 9. This is done by reading off the figure in the 95% column opposite one less than the sample size in the “Degrees of Freedom” column. The figure in the table under 95% and opposite 8 degrees of freedom is 2.31.

Step 4: Multiply the standard error by the t-distribution value i.e. $141 \times 2.31 = 326$

Step 5: Give the 95% confidence limits as $\bar{x} \pm SE = 2500 \pm 326 = 2174$ and 2826 . Thus there is a 95% probability that the mean Saturday morning pedestrian flow outside the supermarket lies between 2174 and 2826. It is worthy noting that if the large sample method previously outlined had been used in this case, the confidence limits would have been calculated as 2234 and 2766, i.e. it would have given a false notion of the accuracy of the estimate.

2.9 Estimates From Sample Counts

So far we have only considered making estimates from samples in which each observation is a measurement, i.e. a number such as pebble roundness, field acreage, or wheat yield, recorded on an interval scale. Often however, the individual observations are counts rather than measurements. For instance, one may wish to determine the proportion of flint pebbles at each end of a beach. In this case each individual observation is recorded on a nominal scale. Random observation is recorded as either “flint” or “not flint”. These can then be counted up, and the proportion of flints in each sample calculated. But one cannot calculate a standard deviation for a single sample of counts, and cannot therefore derive the standard error and confidence limits by the methods outlined previously. One requires a different technique for making population estimates, using the standard error of the binomial distribution.

2.9.1 The Standard Error of the Binomial Distribution

Binomial distribution is associated with the repetition of events where there are only two possible outcomes, e.g. flint or not flint, arable or not arable, immigrants or native etc. If 40% of the pebbles at one end of a beach are flint, then the probability of picking up a flint pebble at random is 0.4, and the probability of picking up non-flint is 0.6. A random sample of 200 pebbles from this part of the beach should yield about $(200 \times 0.4) = 80$ flints and about $(200 \times 0.6) = 120$ non-flints. More generally, if the probabilities of picking up flints and non- flints are p and q respectively (so that $p + q = 1$), then a random sample of size n should yield about np flints and nq non-flints. It is, however, unlikely that two samples

and

Cartography

39

from the same part of the beach would yield exactly the same results. It can therefore be shown that a large number of samples from the same population of pebbles would yield a sampling distribution for the frequency of flint pebbles which would be approximately normal with a mean of np and a standard deviation (called again a standard error) of $\sqrt{npq}$

$$S E np = \sqrt{npq}$$

Provided that the sample size (n) is reasonably large. “Reasonably large” depends upon the value of p and q . If $p = q = 0.5$, the binomial distribution resembles a normal distribution when n is quite small. The greater the difference between p and q the greater

n must be for the distribution to approach normal. A rule – of – thumb is that npq should equal at least 9. Therefore if $p = q = 0.5$, n needs to be at least 36 (i.e. $36 \times 0.5 = 9$); if $p = 0.1$ and $q = 0.9$ (or vice versa) n needs to be at least 100 (i.e. $100 \times 0.1 \times 0.9 = 9$).

The procedure for determining confidence limits from sample counts is therefore as follows.

- (i) Ensure that npq is at least 9, so that the sampling distribution is approximately normal. (if it is not, confidence limits cannot be estimated.)
- (ii) Calculate the mean (np) and standard error ($\sqrt{npq}$) of sampling distribution.
- (iii) Use the z-table to find confidence limits in the same way as for large sample measurements.

Examples

Applied to our 200 pebbles
 Flint pebbles (p) = 40
 %
 Non-flint pebbles (q) = 60% $P = 0.4$, $q = 0.6$, $n = 200$.

Solution

- (i) Therefore mean $np = 200 \times 0.4 = 80$ (ii)

$$\begin{aligned} S E np &= \sqrt{npq} \\ &= \sqrt{200 \times 0.4 \times 0.6} = \sqrt{48} \\ &= 6.93 \approx 7 \end{aligned}$$

- (iii) Confidence limits are 80 ± 7 e.g. 95 % confidence limits are $80 \pm 7 = 66$ and 94

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Cartography

40

99 % confidence limits are $80 \pm 1.58 \times 7 = 62$
and 98.

2.9.2 The Binomial Standard Error as a Proportion.

When making population estimates from counts, one is generally interested in proportions rather than in actual numbers. One does not want to know how many flint pebbles there are on the beach, but what the proportion of flint pebbles is. It is therefore necessary to express the mean and standard error of the sampling distribution as percentages of the sample size (n). Multiplying the numerical values by $100/n$ does this.

Thus, the mean (np) becomes $np \times (100/n) = 100p$ per cent of n and the standard error ($\sqrt{npq}$) becomes $\sqrt{npq} \times \sqrt{100/n}$.

$$= \frac{\sqrt{npqx100^2 / n^2}}{100} = \frac{px100q}{n}$$

This is generally written as $(p\%q\% / n)$, where $p\%$ and $q\%$ express the proportions p and q as percentages (e.g. 0.4 and 0.6 become 40% and 60%). Thus,

$$S E p = \sqrt{\frac{p\%q\%}{n}} \text{ per cent of } n.$$

In the case of the 200 pebbles, the mean for flints becomes $100 \times 0.4 = 40\%$ of the

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Cartography

41

sample and the standard error becomes. $\sqrt{(40 \times 60 / 200)} =$
 $12 = 3.5\% \text{ of } n$

The confidence limits are therefore $40 \pm 3.5 = 33$ per cent and 47 per cent.

We can say with 95% confidence that the proportion of flint pebbles on the part of the beach from which the sample was taken lies between 33 % and 47%.

2.10 Estimating Required Sample Size

Sampling involves a compromise between precision and economy of effort. A parameter can only be obtained with 100 per cent accuracy by measuring or counting the whole population. The smaller the sample, and, to a lesser extent the smaller the sampling fraction, the wider the margin of error of any estimate made from it. In fact, if the sampling fraction is less than one tenth, any variation makes little difference to the margin of error.

If the margin of error that can be tolerated for a particular purpose is known, it is possible to calculate the minimum sample size required, thereby possibly saving time in the collection of unnecessary large sample. Different methods must be used, depending on whether the samples are to be measured or counted, but both involve the prior collection of a pilot sample of about 30 observations.

This must be collected by the same methods that are to be employed in the full sample, and they can be used as part of the full sample, provided they do not affect the selection of additional items.

2.10.1 Sample Size for measured Data.

The formula is:

$$N = (zs/d)^2$$

Where n = required sample size s = std deviation
of pilot sample

d = tolerable margin of error at a specified level of confidence. z = taken
from the z -table corresponding to the same level of confidence.

(The margin of error is the interval between the mean and one confidence limit, and is therefore equal to half the confidence interval).

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Cartography

42

Example

If the pilot sample has a standard deviation of 10, and if the margin of error that can be tolerated is 2.5 at 95 % level, then the required sample size is

$$n = (2 \times 10 / 2.5)^2 = 64$$

Therefore the pilot sample contained 30 observations; an additional 34 would be needed to obtain an estimate within the margin of error that has been stipulated.

2.10.2 Sample Size for Counted Data

This method is used where it is required to estimate the proportion of a population belonging to a specified category, e.g. the proportion of electors who will vote KANU or DP etc.

The formula is

$$N = p\% q\% (z/d)^2$$

Where n, z and d are the same meaning as before and where p and q are the percentages of the pilot sample belonging and not belonging to the specified category respectively. Therefore if 12 out of a pilot sample of 30 said they would vote DP; and if one wished to estimate the proportion of DP votes in the whole electorate with a margin error of 1% either way at the 95% level, then the required size of sample would be

$$N = 40 \times 60 \times (2/1)^2 = 9600.$$

Activity

A pilot sample of householders on an estate showed that income per householder had a standard deviation of \$300 p a, while 21 of the householders were car owners. What size of the sample would be required to estimate;

- (i) The average income per household on the estate, with \$50 margin of error at the 95% level.
- (ii) The proportion of car owners on the estate, with 95% margin of error at the same level.

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Cartography

43

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LESSON 3: HYPOTHESIS TESTING

3.1 Introduction

Much of a geographer's task involves the making of comparisons, both qualitative and quantitative. In most cases Geographers are concerned with the latter, i.e. the statistical comparison of places, areas, and areal distributions. The reason for this preoccupation lies in the nature of Geography as essentially a spatial discipline. Furthermore, comparison often points the way to explanation, which is always an important goal.

Comparison implies the identification of differences, similarities, and associations, which result from two groups of factors:

- (i) Those which operate consistently and from which predictions can be made (e.g. the difference between land use in western province and eastern province of Kenya is partly a function of rainfall. The rainfall is sufficiently predictable to influence the farmers of the two areas to make, in aggregate, different land use decisions.
- (ii) Those which are "irregular, i.e., operate unpredictably and are often characterized as "chance" or "random" factors (e.g. the differences of individual farmers may be partially responsible for deviations from regional norms). The type of statistical comparison to be discussed seeks to distinguish these two groups of factors.

One particular situation, which illustrates the interplay of these two groups of factors, is the sampling process. If exactly 10% of a town's population are pensioners, a random sample of the population can be expected to include about 10% pensioners, but it is unlikely to include exactly 10%. On the other hand, it is very unlikely (though not impossible) that the sample would include 90% pensioners. The consistent factor (that 10% of total population are pensioners), will ensure that about 10% of the total population are pensioners; the inconsistent "chance" factor (the "lack of the draw" in the selection of individuals for the sample), makes it unlikely that exactly 10% will be pensioners.

Now let us reverse the argument. Suppose that random sample of 100 people are taken from towns A and B and that there are 9 and 18 pensioners in the sample from towns A and B respectively. It is unlikely that the proportions of the populations of the two towns who are pensioners are exactly 9% and 18%. The question therefore arises: how precise is "about"? Can we say with any confidence that the former town has a smaller proportion of pensioners than the latter? Or is it likely that the differences in the samples is entirely due to chance i.e. to "sampling error".

3.2 The Null Hypothesis (H_0) A Hypothesis is a "starting point for investigation". A null hypothesis is therefore the starting point of most statistical tests. It is a negative type of proposition, formulated for the purpose of applying a statistical test to a problem under investigation, and generally in anticipation of being rejected as false. In the above examples, we wish to investigate whether there is really a difference in the proportion of pensioners in the two towns, or whether the apparent difference is due to sampling error. There are two possibilities:

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Cartography

45

- (i) that there is a difference in the proportion of pensioners in the two populations; and
- (ii) That there is no difference.

The latter is the negative proportion, and is the **null hypothesis**. More generally, there are two kinds of problems for which different forms of H_0 are appropriate.

- (i) If one is investigating the difference, and, by extension, the similarity between two or more sets of data, then H_0 is of the form. “There is no significant difference between the data sets; which amounts to regarding them as samples from a single background populations, so that any differences between them are no more than might be expected from random variations. This is the main concern of hypothesis testing.
- (ii) If one is investigating the association between variations in data sets (as is the case with problems of correlation), then H_0 is of the form: “There is no association between the variation in corresponding values of the data sets”.

3.3 Procedure in Hypothesis Testing

3.3.1 Using Examples of town A and B

Step 1: Formulate Null Hypothesis (H_0). In the case of towns A and B, it could be stated as “there is no difference in the proportions of pensioners in the two populations.” The point is that H_0 is formulated to calculate the probability that chance alone might yield the given data.

Step 2: Formulate an alternative Hypothesis (H_1). In this case, it could be either that: “There is a difference in the proportion of pensioners in the two towns ”(without specifying which town has the greater proportion); or Town B has a greater proportion of pensioners than town A. H_1 is formulated so that, if H_0 is rejected as a result of the statistical test to be applied, it is logical to adopt H_1 as an alternative.

Step 3: Decide upon a rejection level (α) otherwise called a significant level: If the probability of the difference in the data occurring purely by chance under H_0 is less than α , then H_0 is rejected. The Rejection level (α) is often set at 0.5 or 5% meaning that if differences at least as those of the problem data would occur by chance under H_0 less than five times in a hundred, then H_0 is so unlikely to be true that it is rejected, and the difference between the sets of data is said to be statistically significant at the 0.05 level. Another commonly used rejection level is 0.01, meaning that there is one chance in a hundred that the data being investigated could have occurred under H_0 ; or putting it another way, in the light of the data, there is one chance in a hundred of the null hypothesis being true. The rejection level is a measure of how strong the evidence must be before H_0 is rejected. This

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Cartography

46

should be decided by the investigator before the evidence is examined, i.e. the rejection level should be set before the statistical test is applied; reflecting the risk of a wrong conclusion.

Step 4: Select and carry out an appropriate statistical test to determine the probability (p) that the problem data could have occurred by chance under H_0 .

Step 5: If p (calculated in step 4) is less than α (determined in step 3) then H_0 is rejected at the p – level of significance. If p is greater than α , then H_0 cannot be rejected. This does not mean that H_0 has been proved correct, but simply that the evidence is not sufficiently strong to reject it. A larger sample might provide additional evidence by which H_0 could be rejected.

The procedure outlined above is a useful aid in precise and logical thinking, which should be followed whenever a hypothesis is being tested statistically.

3.3.2 Criteria for the Selection of Statistical Tests

Depends on 3 factors:

- (i) Characteristics of the data
- (ii) The relative “cost – effectiveness of different tests.”
- (iii) Assumptions that can be made about the background population’s from which the data is derived

3.3.2.1 Characteristics of the Data.

Characteristics of the data limit the choice of the test that can be applied to a problem. Among these, the following are especially important.

- (i) They may arise from either counts of items with different qualities or from measurements of different magnitudes.
- (ii) Measurements may be expressed on either an ordinary interval scale.
- (iii) Data may be paired or unpaired.

In addition, one may be comparing a single real data set (or numerical distribution) with a hypothetical distribution; or more than two data sets. Each of these data characteristics has different statistical tests to be applied.

3.3.2.2 The Cost Effectiveness of Different Tests

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Cartography

47

The ‘cost’ in this context means the necessary effort in data collection and calculation. “Effectiveness” means what statisticians call “power”, i.e. the ability to discriminate correctly between true and false hypotheses. Power depends in part on the size of samples being tested, and in part on the power ‘efficiency’ of the test. A test of low power efficiency requires larger samples to achieve the same level of discrimination (power) as a test of high power efficiency. For any given statistical problem there will be

5 one test which is more power efficient than any other that can be applied to it, i.e. it will require smaller samples to achieve a given power than any other tests.

Generally speaking, high powered tests require more carefully measured data, lengthier calculations (unless a computer is available), and make more assumptions about the background population than do low powered test. In choosing between two tests therefore, one has to weigh these disadvantages of the more powerful tests against the fact that the less powerful test will require the collection of larger samples. It is useful in weighing these factors, to know the power efficiency of available tests.

3.3.2.3 Assumptions about the Background Population

Statistical tests may be divided into two families.

- (a) “Classical ” or “Parametric” tests which can only be applied to data on an interval scale.
- (b) “Distribution free” or “non-parametric” tests which can be applied to data on nominal and ordinal as well as interval scales. Parametric tests are generally more powerful, but they make certain assumptions about the background populations from which samples are drawn. The most frequent of these is that the background populations is normally distributed and the smaller the sample being tested, the more nearly normal must the background population be for most parametric tests to be valid. If therefore the normality or near normality of the background population cannot be reasonably assumed then a “non-parametric” test (which makes no assumptions about the distribution of the background population) should be used and this is especially important if the samples being tested are small.

3.4 The X^2 (Chi- Square) Test

The x^2 test is used for comparing counted data, in which individual observations are assigned to categories (i.e. are differentiated on a nominal scale), and the number in each category counted.

X^2 is a simple technique, which works by testing a distribution actually observed in the field against some other distribution determined by the H_0 . For example, where the objects under study are evenly spread over the landscape.

3.4.1 Requirements for the Use of the X^2 test.

- (a) The data must be in the form of frequencies counted in each of a number of categories (percentages cannot be used).

- (b) The total numbers observed must exceed 20
- (c) The expected frequency under H_0 in any one fraction must not normally be less than 5.
- (d) The observations must be independent (i.e. one observation must not influence another).

It is normally not possible to give a meaningful estimate of the power efficiency of χ^2 , because the test is generally used if alternatives are not available or are of doubtful validity.

The formula is simple.

$$\chi^2 = \sum (O - E)^2 / E$$

Where O = the frequencies actually observed. E
= the frequencies expected.

χ^2 is therefore a measure of the aggregate difference between observed frequencies and those expected under H_0 so that the greater its value the less likely it is that H_0 is correct.

Example 1

The problem was to determine whether, in an area of south Devon, the distribution of farms relying mainly on milk production was influenced significantly by differences in surface geology.

Procedure

From the 1-inch to 1-mile geology map equal areas are selected consisting of the following rock types; Keuper Marl, Upper sandstone Upper Marl, Clay - with - Flints, Culm measures. All farms with each area were visited. Any farm obtaining 75% or more of income from milk was classified as mainly relying on milk production. The following data was established.

Area	Rock type	Number of farms relying on milk.
1	Keuper Marl	8
2	Upper Sandstone	10
3	Upper Marl	6
4	Clay-with-flints	3
5	Culm Measures	3
Total		30

Merely by inspection of the above data it might well be inferred that rock type plays a major part in the farmer to produce milk. But before we can draw any conclusion, that is

before we can begin to interpret the evidence we have first to determine the probability that such a distribution could be due to chance. The data are on a nominal scale (obtained by sub-dividing the farms into classes according to the type of rock on which they are situated), and are in form of frequencies. The sample size is small and the total distribution would almost certainly be markedly different from the normal curve. A parametric test is therefore not appropriate, but a test such as X^2 can be used.

Information is also necessary about the size of the farms. A test would not be meaningful if, for example, the 3 farms on the Culm measures were of a larger total area and with high production of milk than the ten farms on the Upper sandstone. In fact this was an area of relatively small farms with no considerable variation in size.

1. Our null hypothesis (H_0), is that the frequency of farms depending mainly on milk for an income is not related to rock type.
2. H_1 states that there is a significant relationship between one rock type and another in the number of farms relying mainly on milk.
3. The rejection level for H_0 was decided at $\alpha 0.05$.
4. The number of farms in each rock type represents the observed frequencies, i.e. the actual numbers counted in the field. If the local geology did not have any influence on the farmers' decision, it would be reasonable to suppose that we should find an approximately even distribution of farms irrespective of rock type. To obtain our "expected" frequencies, therefore, we have to calculate the number of this kind of farms we should expect to find in each area if they were distributed evenly. We have in this case a total of 30 farms in 5 areas of equal size. Our expected frequency in

each area is therefore 6, i.e. $(30/5)$. We now use x^2 to test the probability that the observed frequencies are so different from the expected frequencies that they are unlikely to be due to 'chance' variations. X^2 is a measure of this difference. Obviously the greater the variation between what we 'expect' and what we 'observe' the larger will be the probability that we can reject H_0 .

To find the value of x^2 we use the formula

$$X^2 = \frac{(O-E)^2}{E}$$

Where O = observed frequency

E = expected frequency

Every area over which we have conducted our survey now provides one fraction in the equation thus:

Rock Type	Observed Frequency.	Expected Frequency.	Formula.
Kenper Marl	8	6	$(O-E)^2/E = (8-6)^2 /6$
Upper Sandstone	10	6	$(O-E)^2/E = (10-6)^2/6$
Upper Marl	6	6	$(O-E)^2/E = (6-6)^2/6$
Clay-with-Flints	3	6	$(O-E)^2/E = (3-6)^2 /6$
Calm Measure	3	6	$(O-E)^2/E = (3-6)^2 /6$

X^2 is therefore equal to:

$$\frac{(8-6)^2}{6} + \frac{(10-6)^2}{6} + \frac{(6-6)^2}{6} + \frac{(3-6)^2}{6} + \frac{(3-6)^2}{6}$$

$$= 0.67 + 2.67 + 0 + 1.50 = 6.34.$$

We now have a value for x^2 of about 6.3, which can be referred to the table of critical chisquare values. It will be seen from the table that “degrees of freedom” or ‘df’ has also to be calculated for each test. These are simply one less than the total number of fractions. Thus $df = n - 1$, where n is the number of fraction in the test. In this test we have five fractions. Therefore:

$$Df = 5 - 1 = 4.$$

It is now possible to assess the extent to which the distribution we have been testing may be due to “chance”. Chi-square table shows that with 4 degrees of freedom the critical value of x^2 is 9.49 at the 0.05 level of significance. The calculated value of x^2 in this case is 6.34. Since this is less than the tabled chi-square value at the 0.05 level it indicates that there is a more than 5% probability that the distribution we have tested is random. Or, in other words, one might expect such a distribution to occur purely by chance more than five times in one hundred. We cannot therefore reject the null hypothesis. In spite of the initial impression our conclusion can only be a negative one, that on the basis of the data available there is insufficient evidence to assert that reliance on milk production as a main element of the farm income is affected by rock type.

Note 1; if only two categories are involved (i.e. with one degree of freedom), the expected frequency in any fraction must not be less than 5. If more than two categories are involved (i.e. with $df > 2$) then not more than 20% of the expected frequencies may be less than 5, and in no case must any expected frequency be less than 4.

Note 2: x^2 is used to indicate the value found by calculation from the observed data. The term “chi square ” is reserved for the critical values shown in the chi-square table.

Summary

- 1 Formulate H_0 and H_1 , and decide upon the rejection level.
- 2 Tabulate the observed frequency (O) and calculate the expected frequency (E).

- 3 Using the $\chi^2(O - E)^2/E$, find the value of χ^2
- 3 Obtain the significance of the results the chi-square table of critical values.

Activity

The following data were obtained from the drainage basins of the River Trend.

Area	Units of area	% of total are	No of initial	Expected frequency
A R.Trend	260	17	32	
B R.Dove	355	23	62	
C R Derment	466	30	49	
D R Tame	286	18	51	
E R Soar.	193	12	23	
Totals	<u>1590</u>	<u>100</u>	<u>217</u>	

The problem is this: Are the differences between the basins so great that they are unlikely to be the result of chance? Using the χ^2 test, determine this difference.

So far we have considered the application of χ^2 to one variable (e.g. first farms). The use of the test for two (or more) independent samples is also possible. The example chosen this time also involves measurements on an ordinal scale although for the purposes of this test they are treated as nominal.

Example 2

The problem was to determine whether the angles of natural slopes (having the same aspect) which had developed on Keuper Marl and Greensand in two adjacent areas were significantly different. As the erosional history and vegetation cover of both areas was similar, it may be inferred that significant differences of slope angle result from the nature of the rock itself.

Solution

Measurements were taken randomly in the field using an Abney level with an attempted accuracy of one fifth of degree. The result were then grouped in 5 degrees intervals and tabulated as follows:

	Slope angles	Number of slope measurements	
		Keuper Marl 1	Greensand 2
1	0 - 4.8	21	9
2	5 - 9.8	18	11
3	10 - 14.8	9	15
4	15 - 19.8	8	13
5	20 - 24.8	1	5
		<u>57</u>	<u>53</u>

1. H_0 is that no significant different exists between the natural angles of slope, which have developed upon the two kinds of rock.
2. H_1 is that the differences in slope angle are so great as to make it unlikely that they are the result of “chance” variations.
3. The rejection level is decided as $\alpha = 0.05$.
4. Data is in form of frequencies and χ^2 is chosen to test H_0

The formula is very similar to that used previously.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

The usual symbols used for ‘rows’ and ‘columns’ are r and k . $\sum_r k$ means, “sum over all rows and columns”, (i.e. sum all the cells in the table). The observed frequencies in the above table are substituted for O in the formula. The expected frequency (E) require a little calculation. They are found by multiplying the sum of the row by the sum of the column in which the observed frequencies occurs and dividing by N , the sum of all the observed frequencies. Thus the above table can be rewritten as:

		Keuper Marl		Greensand		$\sum_r$
		O K ₁	E	O K ₂	E	
R ₁	0-4.8	21	15.5	9	14.4	30
R ₂	5-9.8	18	15	11	13.9	29
R ₃	10-14.8	9	12.4	15	11.6	24
R ₄	15-19.8	8	10.9	13	10.1	21

R ₅	20.24.8	1 □	3.1	5	2.1	6
	□k	57		53		N = 110

The observed frequency for $r_1 k_1$ in the above table (the top left-hand value) is 21. The expected frequency appropriate to that number is therefore $(30)(57)/110 = 15.5$; and the fraction which will appear in the formula is $(21 - 15.5)^2/15$. (As 57 out of 110 readings are on Keuper Marl we would expect under H_0 , 57/110 of the total number of slope readings under H_0 to be on Keuper Marl, i.e. 57/110 of 30). It is now possible to substitute actual values in the formula:

$$X^2 = \sum_k \sum_r (O - E)^2 / E$$

$$= (21 - 15.5)^2/15.5 + (9 - 14.4)^2/14.4 + (18 - 15)^2/15 + (11 - 13.9)^2/13.9 + (9 - 12.4)^2/12.4 + (15 - 11.6)^2/11.6 + (8 - 10.9)^2/10.9 + (13 - 10.1)^2/10.1 + (1 - 3.1)^2/3.1 + (5 - 2.9)^2/2.9 = 11.03$$

In order to arrive at a critical value for chi square (i.e. so that it is possible to look up the value in the chi square table), it is necessary to calculate the appropriate degrees of freedom. In this case we take the number of rows less one and multiply the results by the numbers of columns less one or $df = (r-1)(k-1) = (5-1)(2-1) = 4$

With 4 degrees of freedom, the critical values for chi-square are 9.49 at the 0.05 level and 13.28 at the 0.01 level of significance. We can therefore reject H_0 at the 0.05 level, and accept H_1 , that the differences in slope angle are so great as to make it unlikely they are the result of "chance" variation.

3.5 One-Tail and Two – Tailed Tests.

If two random samples are drawn from the same population, the difference between their means will approximate to zero. While the difference between the means of a large number of pairs of samples will tend to be normally distributed about zero, if each of the samples is sufficiently large (say, over 309. Positive differences occur when the first sample mean is greater than the second is greater than the first. Further, if the standard deviation of this sampling distribution (called the standard error of the difference between means) is known, the probability of a particular difference between means occurring under H (ie in samples drawn from the same proportions) can be determined using z-table. This can be illustrated in the figure below.

The figure above depicts a sampling distribution difference-between-means ($\bar{a} - \bar{b}$) with a std. deviation of 4. As with normal distributions 95% of the values (approximately) lie within the range of two std. Deviations of the mean i.e. within -8 and $+8$, since the mean is zero. Therefore the probability of obtaining two sample means which is outside this range is 5% or 0.05 and is represented by the shaded areas under both extremities or “trails” of the curve, added together; z-tables give probabilities associated with values lying outside other ranges.

The shaded area in col. B of Appendix 1 (probabilities associated with values of z in a Nominal distribution) is at one extremity of the normal curve.

In column A, the shaded area is adjacent to the mean.

This is because Column A is used to calculate confidence limits, and for this purpose we need to know the probabilities associated with values lying within different distances of the mean of the sampling distribution; while column B (and col c) are used for testing hypothesis when we want to know the probabilities associated with values lying beyond different distances of the mean of a sampling distribution. Therefore when calculating confidence limits with z-tables, Appendix 1A is used and probabilities (confidence levels) are usually of the order of 95% or 99% and when hypotheses are tested with the aid of z-tables, Appendix 1B or C is used, and probabilities (significance levels) are of the order of 5% or percent more generally expressed as 0.05 or 0.01 of course, not all hypothesis tests make use of z-tables, but the same principle applies with other sampling distributions.

When H_1 is of the type “there is no significant difference between the data” the alternative hypothesis (H) can take one of two forms.

- (a) non-directional – in which only fact of there being a difference between the population represented by the data is stated
- (b) Directional – in which it is specified which population has the greater value of whatever the data represented.

Referring to the figure above, if H is non-directional. It is not specifying whether $\bar{a} - \bar{b}$ is expected to be positive or negative. A difference between $\bar{a}$ and $\bar{b}$ can be due to $\bar{a}$

exceeding b (so that $a - b$ is positive), or to b exceeding a (so that $a - b$ is negative). The former case is represented by the right hand half of the curve, the latter by the left. Thus, a difference of more than 8 which does not specify whether a is greater-or less

than b , is as we have seen represented by the area under both tails (the shaded areas in the figure). Which together constitute 5% of the total area under the curve, (2.5 under each tail), indicating a 5% or 0.05 probability of occurrence. On the other hand, if it is

directional, it is specifying whether $a - b$ is expected to be positive or negative, and therefore also the tail of the sampling distribution in which the value $a - b$ is expected to lie. Thus, if H_1 states that a exceeds (rather than simply differs from) b , we are only

interested in the right hand half of the curve, the probability that a will exceed b by more than 8 is represented by the shaded area under the right-hand tail only. This constitutes only 2.5% of the total area under 0.025.

Similarly if H_1 states b to be greater than a , then the probability of that difference being greater than 8 is represented by the shaded area under the left-hand tail of the curve (($a -$

b) is less than 8, i.e. further from zero than -8 . This is also 2.5% i.e. 0.025. Therefore the probabilities associated with a directional alternative hypothesis are 'one-tailed' and only half the "two-tailed" probabilities associated with a non-directional alternative hypothesis.

It follows that where H_1 is directional a one-tailed test is called for? Where H_1 is non-directional, two-tailed test is required. Therefore the nature of the alternative hypothesis determine whether a test should be one-or two-tailed, and this may determine whether or not a null hypothesis is to be rejected. An alternative hypothesis, requiring a one-tailed test is statistically less demanding (i.e. H_0 is more easily rejected) and should only be formulated if there are strong reasons, external to the data being tested. If in doubt it is wise to formulate a non-directional alternative hypothesis and apply a two-tailed test. In the last resort, the choice between a directional and non-directional alternative hypothesis is, like the selection of the rejection level (α), arbitrary; it is based on common sense and experience rather than on mathematical logic.

3.6 The t -Test.

The t-test makes use of the t-distribution. It is used to determine the significance of the difference between two groups of data measured on an interval scale.

3.6.1 The t-test for Independent Samples

Sample values of two independent variables, x and y, together with deviations from sample means $(x - \bar{x})$ and $(y - \bar{y})$, and the squares of these deviations are given in the table below

x	y	$(x - \bar{x})$	$(x - \bar{x})^2$	$(y - \bar{y})$	$(y - \bar{y})^2$
10	8	3	9	2	4
8	7	1	1	1	1
7	7	0	0	1	1
6	6	-1	1	0	0
6	5	-1	1	-1	1
5	3	-2	4	-3	9
$\bar{x} = 7$	$\bar{y} = 6$		$\Sigma(x - \bar{x})^2 = 16$		$\Sigma(y - \bar{y})^2$

$N_x = 6 = n_y$ = size of each sample (but samples do not need to be equal in size for a t-test for independent sample). Before the t-test can be applied, two assumptions must be made:

- (i) The background population of the samples are approximately normally distributed. (ii)
- (iii) The standard deviations of the population from which the samples are drawn are equal.

Checks to determine these assumptions should be carried out before the t-test is used.

3.6.2 Steps in Hypothesis testing

1. Null hypothesis (H_0): there is no difference between the means of the populations of which x and y are samples.
2. Alternative hypothesis (H_1) there is a difference between the means of the populations of which x and y are samples.
3. Rejection level (α : 0.05 4. The t-test.

- (a) Calculate the standard deviations of the populations. Because of assumptions
- (b) Above, and because the samples are so small, a pooled best estimate (Bissell's correction) ($\bar{s}$) is obtained from all the data in both samples from the formula.

$$\bar{s} = \sqrt{\frac{\Sigma(x - \bar{x})^2 + \Sigma(y - \bar{y})^2}{n_x + n_y - 2}}$$

$$\sqrt{16 \square 16 / 6 \square 6 \square} = \underline{\quad} \quad \underline{3.2}$$

$$\square \square$$

(b) Calculate the standard deviation of the sampling distribution of

$$\begin{array}{l} \text{S E } x \\ 0.53 \end{array} \quad \begin{array}{l} \sqrt{n_x} = \sqrt{3.2 / \sqrt{}} \\ \sqrt{n_v} = \sqrt{3.2 / \sqrt{}} \end{array} \quad \square \quad = /6 = 0.53 \quad \square \quad = 0.73 \quad \text{S E } y = /6 = \\ = 0.73 \quad \underline{\quad} \quad \underline{\quad}$$

(b) Calculate the standard deviation of the sampling distribution of the difference

– between- means (i e the standard error of $\bar{x} - \bar{y}$). This is obtained, using formula.

$$\begin{array}{c} \bar{x} \quad \bar{y} \\ \text{S-E } (\bar{x} - \bar{y}) = \\ (\text{S E } x)^2 + (\text{S E } y)^2 \end{array} \quad \sqrt{\quad}$$

Therefore $\text{SE } (\bar{x} - \bar{y}) = (0.53 + 0.53) = 1.06 = 1.03$

(c) Calculate t.

T = the difference between the means the standard errors of the difference.

$$T = \bar{x} - \bar{y} / \text{S E } (\bar{x} - \bar{y}) = 7 - 6 / 1.03 = 1 / 1.03 = 1$$

(e) as the rejection level is 0.05, enter the t-table under the column-heading 0.05, and along the row for $(n_x + n_v - 2)$ degrees of freedom ie $(12 - 2) = 10$ degrees of freedom. This gives a critical value of 2.23.

for the difference between sample means $(\bar{x} - \bar{y})$ to be significant, t must be more than 2.23 (note: there are two-tailed critical values of t, which is appropriate for this example, because H_1 was non-directional. If H_1 had been directional, the critical value of t would have been found under the 0.10 column heading, since a two-tailed probability of 0.10 is equal to one-tailed probability of 0.05. the critical value would then have been 1.81)

The decision to reject or retain H_0 . For H_0 to be rejected, t must be more than the critical value found in the t -table. In this case t was calculated to be 1, the critical value was

found to be 2.23. Therefore H_1 cannot be rejected. The difference between $\bar{x}$ and $\bar{y}$ is not significant at the 0.05 level.

3.6.3 The t-test for Paired Samples

The same values of x and y are used to illustrate this method, but because individual values of x and y are paired, the differences between them (d) are used. Thus

x	y	d	$(d - \bar{d})$	$(d - \bar{d})^2$
10	8	2	1	1
8	7	1	0	0
7	7	0	-1	1
6	6	0	-1	1
6	5	1	0	0
5	3	2	1	1
$\bar{x} = 7$	$\bar{y} = 6$	$\bar{d} = 1$	$\sum (d - \bar{d})^2 = 4$	

The assumptions and steps 1 to 3 outlined t-test for independent samples apply again here. The difference in procedure comes in step 4, the test.

- (a) Calculate the best estimate of the standard deviation of the difference (d) between paired values of x and y , using the following formula.

$$s_d = \sqrt{\frac{\sum (d - \bar{d})^2}{n-1}} = \sqrt{\frac{4}{5}} = 0.8$$

- (b) Calculate the standard deviation of the sampling distribution of mean differences (ie the standard error of mean difference).

$$S E \bar{d} = \frac{s_d}{\sqrt{n}} = \frac{0.8}{\sqrt{6}} = \frac{0.8}{2.45} = 0.33 = 0.36$$

- (c) calculate t t = the difference between the means the standard error of the difference

$$t = \frac{\bar{x} - \bar{y}}{S E \bar{d}} = \frac{1}{0.36} = 2.8$$

- (d) As the rejection level is again 0.05, and as H is again non-directional, enter the tables (which give two-tailed probabilities in the column headed 0.05 and opposite $(n-1)$ degrees of freedom i.e. opposite 5 degrees of freedom. The critical value of t is seen to be 2.57. steps 5. The decision to reject, or retain H_0 . For H_0 to be rejected, t must be more than the critical value found in the t -table. In this case, t was calculated to be 2.8; the

critical value of t is 2.57, therefore H_0 can be rejected; the difference between x and y is significant at the 0.05 level.

This if values of x and y are independent as in the first example, their means are not significantly different; if the same values are paired as in the last example their means are significantly different.

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